Lab #4: DHCP Starvation - Write-up

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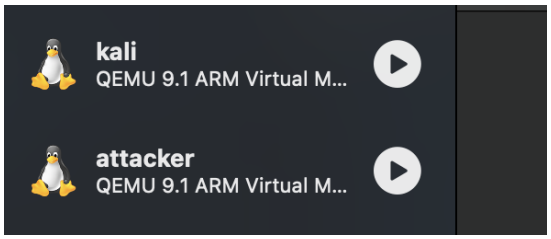
This lab aimed to demonstrate the vulnerability of a DHCP server to a DHCP Starvation attack. The objective of this attack is to exhaust the pool of available IP addresses on a DHCP server by flooding it with forged IP requests, thereby preventing legitimate clients from obtaining an IP address and connecting to the network. During the lab, we set up a dedicated DHCP server and developed a custom Python tool to perform the attack

Lab Setup

Environment: Two virtual machines on VirtualBox

- **Machine 1: Attacker & DHCP Server (Kali Linux)**
- **Machine 2: Victim (Kali Linux)**

Network Mode: Internal Network (VirtualBox), ensuring isolated communication between the VMs.



We installed the ISC DHCP server using the command: sudo apt-get install isc-dhcp-server

We configured The eth0 network interface of the Kali machine with a static IP address within the designated network range (172.16.15.0/24), which would serve as the DHCP server's IP address.

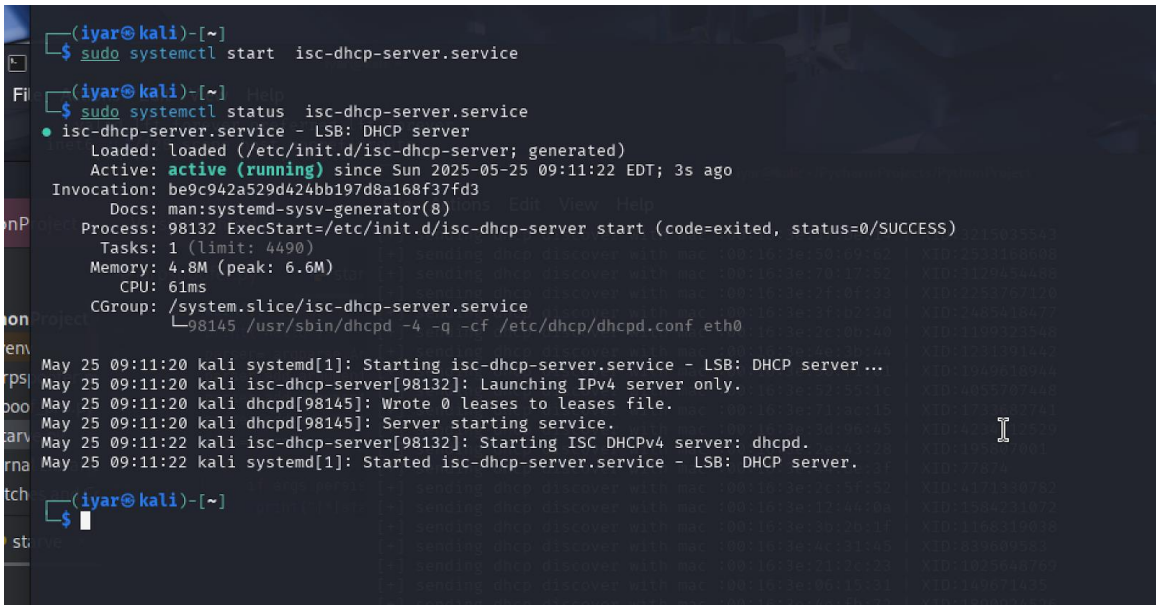
/etc/dhcp/dhcpd.conf contains the main DHCP server settings. We defined the IP address range for allocation, a default gateway, and DNS servers.

we encountered a problem with /var/lib/dhcp/dhcpd.leases

If the file doesn't exist or the server lacks write permissions, the server will fail to start or operate correctly.

so in solution We manually created the file and set appropriate permissions:

After all configurations and fixes, we restarted the DHCP service and checked its status: The goal was to see Active: and connect the second virtual machine (the client) to the network and verify that it received an IP address within the 172.16.15.x range before launching the attack.



Attack Description

The tool was developed in Python using the **Scapy** library for crafting and sending DHCP packets. Its goal is to **flood a DHCP server with DHCP Discover and Request packets** using randomized MAC addresses, ultimately exhausting the pool of available IP addresses.

Key Functions and Logic:

get_random_mac()
Generates a random MAC address using a fixed prefix (00:16:3e) and randomized suffix to simulate multiple clients from a VirtualBox-like environment.

mac2str(mac)
Converts a MAC address from string format ("00:11:22:33:44:55") to the byte format required by the BOOTP chaddr field.

Packet Construction:

send_dhcp_discover(mac, xid, iface)
Constructs and sends a DHCP Discover packet:

- Ether(src=mac, dst="ff:ff:ff:ff:ff:ff"): Broadcasts the packet on the Ethernet layer.
- IP(src="0.0.0.0", dst="255.255.255.255"): Used for DHCP Discovery.
- UDP(sport=68, dport=67): Standard DHCP client-server ports.

- BOOTP(chaddr=mac2str(mac), xid=xid): Specifies the client's MAC and transaction ID.
- DHCP(options=[("message-type", "discover"), "end"]): Marks this as a Discover message.

send_dhcp_request(mac, xid, offered_ip, server_ip, iface)
Sends a DHCP Request message based on the offer received:

- Requests the offered IP and identifies the server.
- Helps complete the DHCP handshake to reserve the address.

Listening for DHCPOFFER:

dhcp_offer_filter(pkt, xid)
Filters incoming packets to identify DHCP Offer responses that match the transaction ID (XID) of the request.

Main Attack Logic – dhcp_starvation():

- Loops (once or persistently) to:
    1. Generate a new MAC and transaction ID.
    2. Send a DHCP Discover packet.
    3. Sniff for a DHCPOFFER response (up to 3 seconds).
    4. If offer received, extract offered IP and server IP.
    5. Send a DHCP Request to reserve the offered address.
- If --persist is used, the loop continues endlessly, slowly exhausting all available IPs.

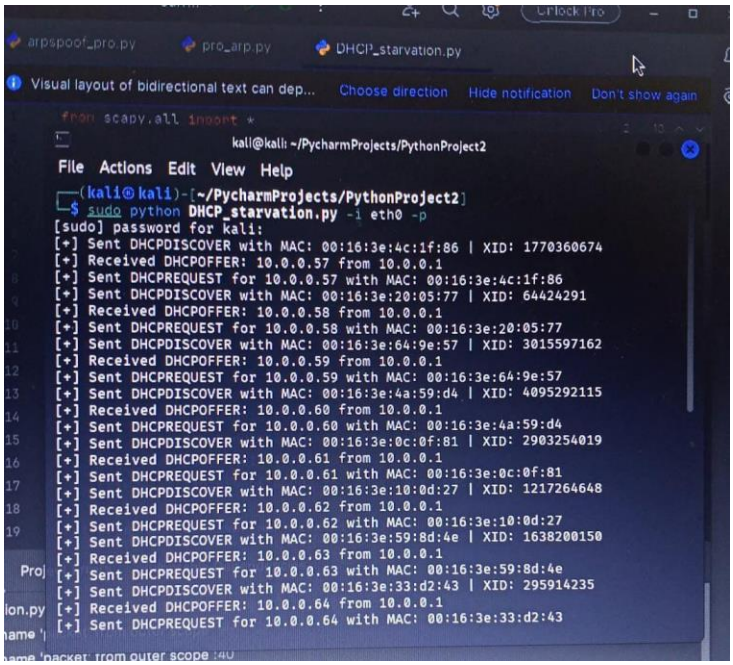
Bonus Feature – Persistent Mode:

Added support for -p / --persist, which keeps the attack running continuously. This allows for **sustained denial-of-service**, blocking new clients from receiving IPs even after lease expirations.

Graceful termination with Ctrl+C is handled via a try/except KeyboardInterrupt.

Running the Attack Script

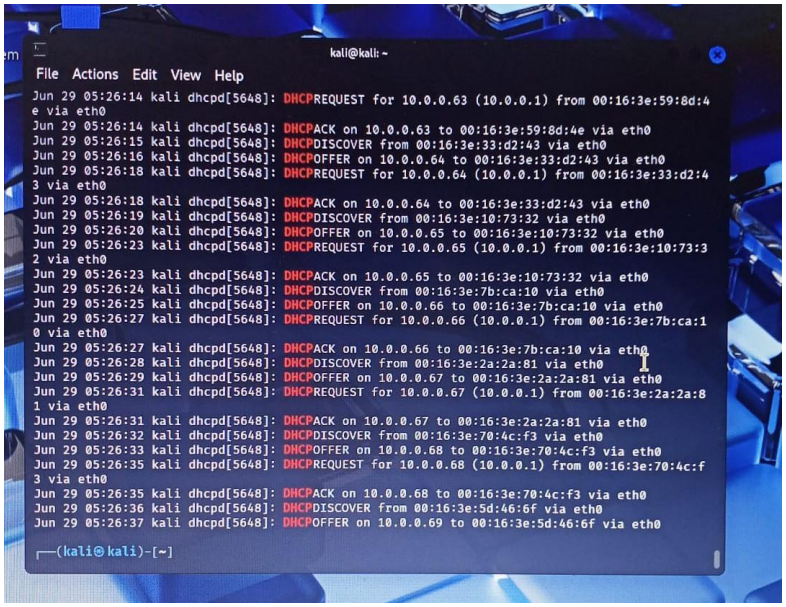
This image shows the execution of the DHCP Starvation script. The script sends DHCPDISCOVER and DHCPREQUEST packets with random MAC addresses in order to exhaust the DHCP server's IP address pool.



Server Logs via journalctl

This image shows the output of the following command:

journalctl -u isc-dhcp-server | grep DHCP

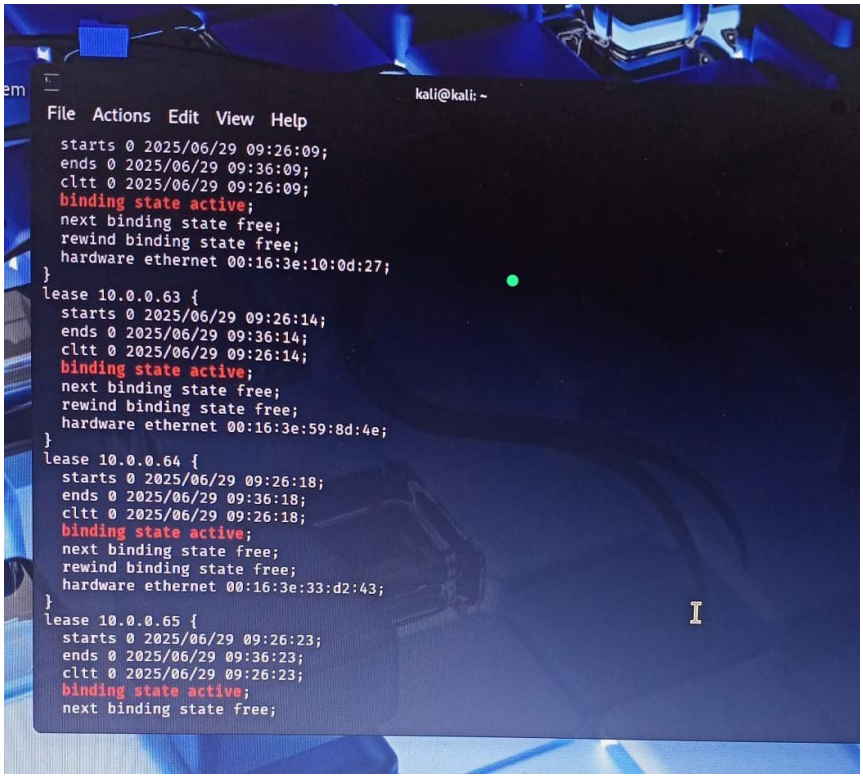


The output confirms that the server received and processed the requests sent by the script, showing the assigned IP addresses.

Inspecting the dhcpd.leases File

This image shows the output of the command:

cat /var/lib/dhcp/dhcpd.leases | grep -A 10 "binding state active"



The output displays active lease entries, indicating that the IP addresses were successfully allocated and recorded by the server, each with a unique MAC address.

This lab successfully demonstrated the setup of a DHCP server, the development of a tool to perform a DHCP Starvation attack, and the verification of the attack's impact on both the DHCP server and clients on the network. The development process involved overcoming several configuration and syntax challenges, the resolution of which sharpened our understanding of network protocols and security tools. The implementation of the bonus challenge added a "persistence" capability to the tool, making it more effective in sustained denial-of-service scenarios.